

Linear algebra and geometry 1 - TTPU

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Marks -35.00/1000.00

Grade -1.05 out of 30.00 (-4%)

Question 1

Incorrect

Mark -15.00 out of 100.00

Let $\vec{i}, \vec{j}, \vec{k}$ the versors of the coordinate axes of $\mathbb{R}^3$. For every $a \in \mathbb{R}$, let us consider the vectors

$$\vec{u} = -2\vec{j} + a\vec{k}, \quad \vec{v} = 2\vec{i} + \vec{k}, \quad \vec{w} = -a\vec{i} - \vec{j}.$$

Which of the following statements is true?

- ☒ (a) There exists a unique $a \in \mathbb{R}$ such that the vectors $\vec{u}, \vec{v}, \vec{w}$ are linearly independent. ✗
- ☐ (b) For every $a \in \mathbb{R}$ the vectors $\vec{u}, \vec{v}, \vec{w}$ are coplanar.
- ☐ (c) For every $a \in \mathbb{R}$ the vectors $\vec{u}, \vec{v}, \vec{w}$ are linearly independent.
- ☐ (d) There exists a unique $a \in \mathbb{R}$ such that the vectors $\vec{u}, \vec{v}, \vec{w}$ are coplanar.

Risposta errata.

The correct answer is: For every $a \in \mathbb{R}$ the vectors $\vec{u}, \vec{v}, \vec{w}$ are coplanar.

Question 2

Incorrect

Mark -15.00 out of
100.00

Given the straight lines

$$r : \begin{cases} 2x - 5y = 7 \\ 2z - 3y = 5 \end{cases}, \quad \text{and} \quad s : \begin{cases} x - y - z = 1 \\ x - 4y + z = 5, \end{cases}$$

find the right statement.

- ☒ (a) The lines are skew. ✗
- ☐ (b) The lines intersect at exactly one point.
- ☐ (c) The lines are parallel and distinct.
- ☐ (d) None of the other statements is correct.

Risposta errata.

The correct answer is: The lines are parallel and distinct.

Question 3

Incorrect

Mark -15.00 out of
100.00If the linear system $AX = B$ with 4 equations and 6 variables has ∞^m solutions with $m \leq 4$, then necessarily

- ☒ (a) $\text{rank}(A) = 2$. ✗
- ☐ (b) $\text{rank}(A) \geq 2$.
- ☐ (c) $\text{rank}(A|B) = 2$.
- ☐ (d) $\text{rank}(A) \leq 2$.

Risposta errata.

The correct answer is: $\text{rank}(A) \geq 2$.

Question 4

Incorrect

Mark -15.00 out of
100.00

If

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix},$$

then

-
- ☒ (a) $\mathbf{A}^{-1} = \begin{bmatrix} -1 & -1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ ✖
- ☐ (b) $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$
- ☐ (c) $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$
- ☐ (d) $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

Risposta errata.

The correct answer is: $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

Question 5

Incorrect

Mark -15.00 out of
100.00

Let $f : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be an endomorphism with characteristic polynomial $P(t) = (t^2 - 2)(t^2 - t)$.

Which one of the following statements is true?

-
- ☒ (a) $\dim(\text{Ker}(f)) = 2$. ✖
- ☐ (b) $\dim(\text{Im}(f)) = 1$.
- ☐ (c) $\dim(\text{Im}(f)) = 2$.
- ☐ (d) $\dim(\text{Ker}(f)) = 1$.

Risposta errata.

The correct answer is: $\dim(\text{Ker}(f)) = 1$.

Question 6

Incorrect

Mark -15.00 out of
100.00

In $\mathbb{R}^4$ let us consider the vector subspace

$$V = \{ (x, y, z, w) \mid x + y - z - w = 0, x - 2y = 0 \}.$$

Which one of the following statements is true?

- ☒ (a) V is the vector subspace generated by the vectors $(1, 1, 1, 1)$ e $(0, 0, 1, 1)$.
✗
- ☐ (b) A basis of V is formed by 3 vectors.
- ☐ (c) V is the vector subspace generated by the vectors $(2, 1, 2, 1)$ e $(0, 0, 1, -1)$.
- ☐ (d) V is a subspace of dimension 1.

Risposta errata.

The correct answer is: V is the vector subspace generated by the vectors $(2, 1, 2, 1)$ e $(0, 0, 1, -1)$.

Question 7

Incorrect

Mark -15.00 out of
100.00

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the endomorphism $f(x, y) = (3x - y, 3y)$.

Which one of the following statements is true?

- ☒ (a) f is simple. **✗**
- ☐ (b) f has two distinct eigenvalues.
- ☐ (c) $(2, 0)$ is an eigenvector.
- ☐ (d) $(0, 1)$ is an eigenvector.

Risposta errata.

The correct answer is: $(2, 0)$ is an eigenvector.

Question 8

Incorrect

Mark -15.00 out of
100.00

In the euclidean 2-dimensional space with a fixed cartesian system, consider the conic $\mathcal{C}$ having equation

$$x^2 - 6xy + 9y^2 + 2x - 6y + 1 = 0.$$

Which of the following statements is true?

- ☒ (a) $\mathcal{C}$ is a not degenerate ellipse. ✖
- ☐ (b) $\mathcal{C}$ is degenerate.
- ☐ (c) $\mathcal{C}$ is a not degenerate hyperbola.
- ☐ (d) $\mathcal{C}$ is a not degenerate parabola.

Risposta errata.

The correct answer is: $\mathcal{C}$ is degenerate.

Question 9

Incorrect

Mark -15.00 out of
100.00

In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the circumferences of equations

$$\mathcal{C}_1 : x^2 + y^2 + z^2 - 1 = x = 0, \quad \mathcal{C}_2 : x^2 + y^2 + z^2 - 1 = x - z = 0,$$

having radius respectively R_1 and R_2 .

Which of the following statements is true?

- ☒ (a) $R_1 < R_2$. ✖
- ☐ (b) $\mathcal{C}_1$ consists of only one point.
- ☐ (c) $R_1 = R_2$.
- ☐ (d) $R_1 > R_2$.

Risposta errata.

The correct answer is: $R_1 = R_2$.

Question 10

Correct

Mark 100.00 out of
100.00

Let us consider the quadratic form defined by

$$q(x, y) = (x, y)A \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{with} \quad A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

Which of the following statements is true?

-
- ☒ (a) $q(a, b) \geq 0$ for every $(a, b) \in \mathbb{R}^2$. ✓
 - ☐ (b) The matrix A has two negative eigenvalues.
 - ☐ (c) There exists $(a, b) \in \mathbb{R}^2$ such that $q(a, b) < 0$.
 - ☐ (d) The columns of A are eigenvectors for A .

Risposta corretta.

The correct answer is: $q(a, b) \geq 0$ for every $(a, b) \in \mathbb{R}^2$.